



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 103



AGRICOMP – PhilMech Cassava Belt Dryer

AGRICOMP – PHILMECH CASSAVA BELT DRYER

Test Report No.	:	2013 - 103
Date of Test	:	May 24, 2013
Test Valid Until	:	May 24, 2018
Brand	:	AGRICOMP - PHilMech
Model	:	-
Type	:	Continuous flow type with direct-fired biomass-fed furnace
Manufacturer	:	AGRICOMPONENT CORPORATION National Hi-Way, Tagaran Cauayan City
Test Requested by:	:	AGRICOMP MACHINERIES & CONSTRUCTION CORPORATION Tagaran, Cauayan City
Machine Classification	:	Commercial

SUMMARY

The AGRICOMP – PHilMech Cassava Belt Dryer with AMCC Big Ben direct-fired biomass-fed furnace was tested for performance using granulated cassava as test material. It had an Input Capacity of 893.4 kg/h and an Output Feed Rate of 395.2 kg/h with corresponding Product Recovery of 44.2% from an initial Moisture Content of 51.0%_{w.b.} to a Final Moisture Content of 10.8%_{w.b.} with an average Moisture Reduction per Hour of 40.2%. The average Drying Air Temperature during the test was 97.9°C. The Fuel Consumption of the direct-fired biomass furnace using corn cob as fuel was 125.9 kg/h while the Total Power Consumption of the

electric motors which drove the components of the dryer was 11.9 kW with Total Load Current of 41.5 amperes and Line Voltage of 211 volts.

The Heating System Efficiency of the dryer was 71.4%. Heat Utilization was 3896 kJ per kg moisture removed with a corresponding Drying Efficiency of 60.0%. The dryer had a Drying System Efficiency of 42.9%.

PURPOSE AND SCOPE OF TEST

The AGRICOMP – PHilMech Continuous Flow Cassava Belt Dryer coupled to an AMCC Big Ben direct-fired biomass-fed furnace was tested for drying granulated cassava to determine its operating performance. The test was conducted to verify and assess the following:

- 1.0 Drying System Performance
 - 1.1 Input Capacity
 - 1.2 Output Feed Rate
 - 1.3 Heat Utilization
 - 1.4 Drying Efficiency
 - 1.5 Drying System Efficiency
 - 1.6 Power Consumption
- 2.0 Heating System
 - 2.1 Heating System Efficiency
 - 2.2 Furnace Fuel Consumption Rate (using corn cob)

PLACE AND DATE OF TEST

The AGRICOMP – PHilMech Cassava Belt Dryer was tested in Brgy. Villa Luna, Cauayan City, Isabela on May 24, 2013.

METHODS OF TEST

The AGRICOMP – PHilMech Cassava Belt Dryer was tested based on the Philippine Agricultural Engineering Standard “PAES 202:2000 Agricultural Machinery – Heated-Air Mechanical Grain Dryer - Methods of Test”.

OPERATOR’S MANUAL

No operator’s manual or brochure was submitted.

DESCRIPTION OF THE GRAIN DRYER

Figure 1 shows the AGRICOMP - PHilMech Cassava Belt Dryer coupled to a direct-fired AMCC Big Ben Biomass Furnace. It was a continuous flow type dryer designed for drying granulated cassava. It consisted of the following main components: grain loading system, drying system, control panel and biomass furnace.

LOADING SYSTEM

The loading system consisted of a loading hopper (**Figure 2**), a bucket elevator (**Figure 3**), feed hopper with screw conveyor (**Figure 4a**) and feed belt (**Figure 4b**). It was used for loading granulated cassava to the belt conveyors. The loading hopper was located on the lower portion of the elevator at ground level. The elevator was a belt-bucket centrifugal discharge type.

DRYING SYSTEM

The drying system consisted of a drying chamber, three hot-air fans and an unloading system.

Drying Chamber

This was where the granulated fresh cassava were being dried. It consisted of five belt conveyors (**Figure 5**) made of No. 8 mesh stainless screen. A thin layer of granulated cassava was conveyed along the first belt and discharged from the opposite end of the first belt conveyor to the intake end of the second belt conveyor underneath. The second belt conveyor moved in opposite direction relative to the first conveyor. The process of conveying the thin layers of granulated cassava along the belt and discharging to the next belt conveyor was repeated until the bottommost belt conveyor was reached. The dried granulated cassava was discharged from the bottommost belt to the discharge auger (**Figure 6**).

Hot air Fans. (Figure 7)

The dryer was provided with three hot air fans of axial type. These were driven by separate electric motors through belts and pulleys. These were installed on top of the drying chamber. The fans sucked the hot air that passed through the granulated cassava. The exhaust air was conveyed outside through air ducts.

CONTROL PANEL (Figure 8)

The dryer control system was housed in a rectangular box compartment. It included the voltmeter, switches and controls of the dryer.

BIOMASS FURNACE (Figure 9)

The dryer was coupled to the AMCC Big Ben Biomass Furnace which was a direct-fired furnace and which used either corn cobs or rice hull as fuel. The combustion chamber (Figure 10) was made of M. S. plate, cylindrical in shape and lined with fire bricks where combustion took place. It was enclosed by a rectangular box made of M. S. plate. Fuel was fed through an auger installed at the bottom of the hopper. A centrifugal fan provided primary air to the burning chamber through holes in the brick wall. Ash fell beneath the burning chamber and was collected manually. Ambient air was heated when it entered the biomass furnace and was sucked by an intake blower (Figure 11) in the air duct leading to the dryer.



Figure 1. AGRICOMP – PHilMech Cassava Dryer



Figure 2. Loading hopper



Figure 3. Bucket Elevator

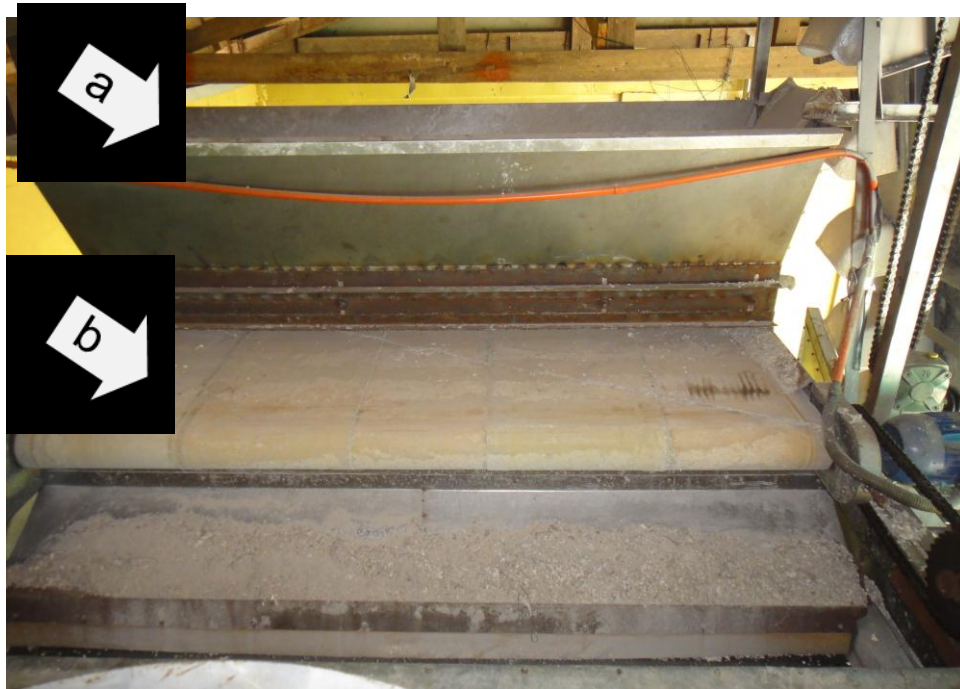


Figure 4. a) Feed hopper with screw conveyor
b) Feed belt

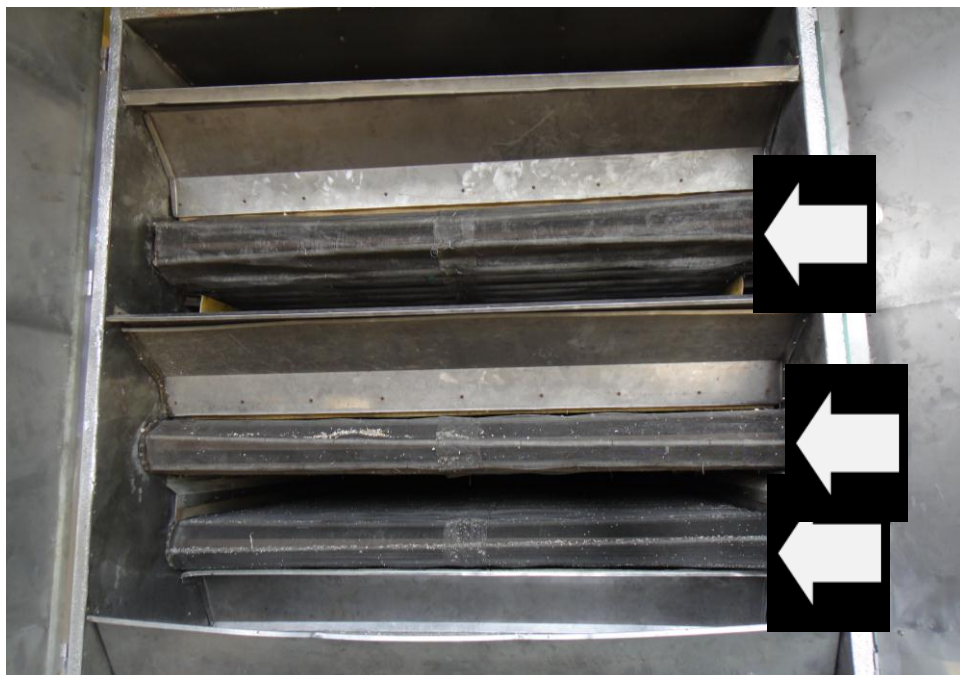


Figure 5. Belt conveyors



Figure 6. Discharge auger



Figure 7. Hot air fan



Figure 8. Control Panel

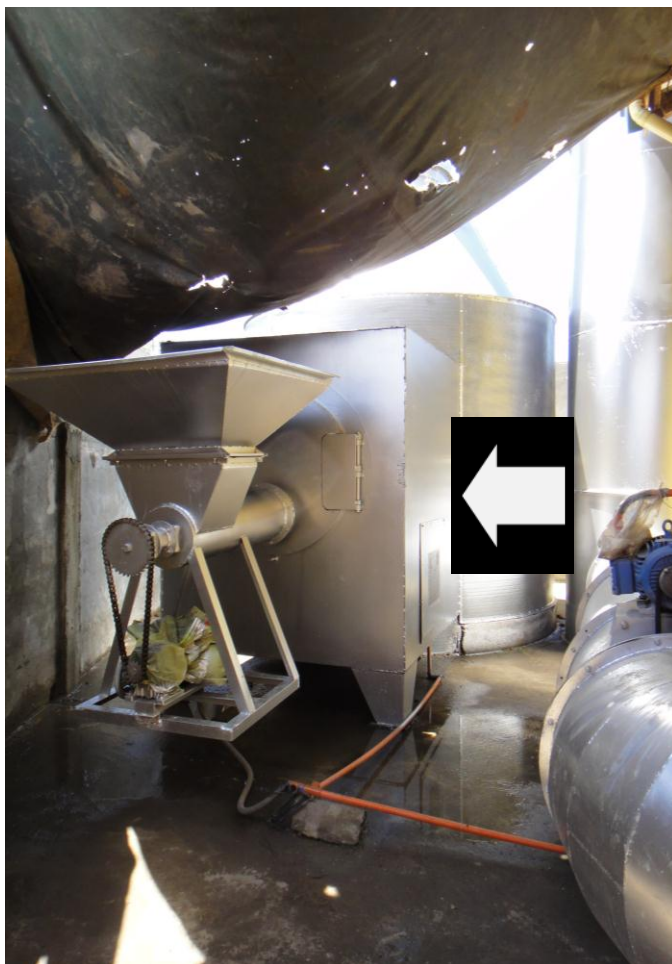


Figure 9. Biomass Furnace



Figure 10. Combustion Chamber

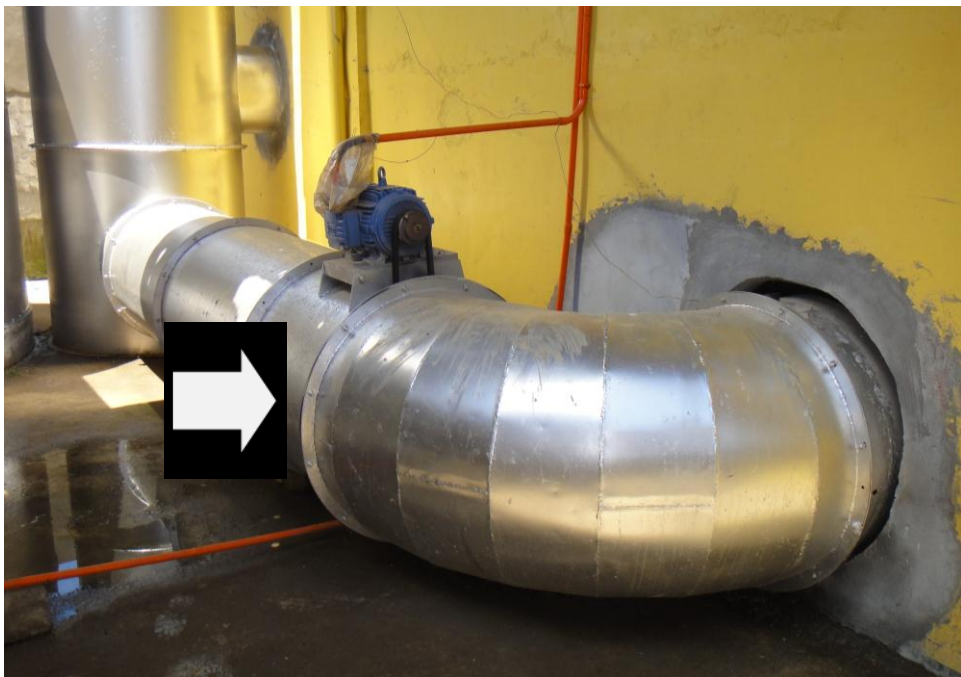


Figure 11. Intake Blower

RESULTS AND DISCUSSION

The AGRICOMP – PHilMech Cassava Belt Dryer with AMCC Big Ben direct-fired biomass-fed furnace was tested for performance using granulated cassava as test material.

Results of performance tests are shown in **Table 1**.

Input Capacity

The input capacity of the dryer was the amount of fresh granulated cassava fed to the drier per unit of time. The dryer had an Input Capacity of 893.4 kg/h

Output Feed Rate

The output feed rate was the amount of dried granulated cassava output of the drier per unit of time. The drier had an Output Feed Rate of 395.2 kg/h.

Product Recovery

The product recovery was the ratio of the amount of dried granulated cassava to the amount of fresh granulated cassava fed to the drier per unit of time, expressed in percent. The drier had a Product Recovery of 44.2%.

Heating System Efficiency

Heating system efficiency was the ratio of the amount of heat supplied for drying to the total amount of heat available from the fuel, expressed in percent. The dryer had Heating System Efficiency of 71.4%.

Furnace's Fuel Consumption

To determine the fuel consumption of the AMCC biomass furnace, the total amount of corn cob consumed during the test was divided by the total time of operation. The Fuel Consumption of the furnace was 125.9 kg/h of corn cob.

Heat Utilization

Heat utilization referred to the total amount of heat utilized to vaporize moisture from the grain. Results showed that the dryer had a Heat Utilization of 3896 kJ/kg of water removed.

Drying Efficiency

Drying efficiency was the ratio of the total heat utilized to vaporize moisture in the granulated cassava to the amount of heat added to the drying air expressed in percent. Results showed that the dryer had a Drying Efficiency of 60.0%

Drying System Efficiency

Drying system efficiency was the ratio of the total heat utilized for drying to the heat available in the fuel expressed in percent. Results showed that the dryer had a Drying System Efficiency of 42.9%.

Power Consumption

The average total Power Consumption of the dryer during operation was 11.9 kW with Load Current of 41.5 amperes and Line Voltage of 211 volts.

Labor Requirement

Three persons (**Figure 12**) were needed to operate the dryer i.e. one to replenish fuel to the furnace, during the drying operation and two for loading of fresh granulated cassava to the drier.

Observations

The dryer operated smoothly and no breakdown or failure of components occurred during the test. However, it was observed that during loading operation, some fresh granulated cassava fall to the ground from the discharge chute of the elevator (**Figure 13**).



Figure 12. Operators of the drier



Figure 13. Granulated cassava that fall to the ground during loading

PERFORMANCE TEST

Machine Tested : AGRICOMP- PHilMech Cassava Belt Dryer

Date of Test : May 24, 2013

Location of Test : Brgy. Villa Luna, Cauayan City

Table 1. Conditions of Test Materials

ITEMS	Data
1. Crop	Cassava
2. Variety	Unknown
3. Condition	Granulated
3. Moisture Content, $MC_{w.b.}$ (%)	51.0
4. Bulk Density (kg/m^3)	668.3

Table 2. Results of Performance Test

ITEMS	DATA
1. Initial Weight of Test Material (kg)	3797
2. Input Time (h)	4.25
3. Input Capacity (kg/h)	893.4
4. Output Feed Rate (kg/h)	395.2
5. Product Recovery (%)	44.2
6. Final Moisture Content (%)	10.8
7. Rate of Moisture Removal (kg/h)	498.2
8. Moisture Reduction Rate (%/h)	40.2

Table 2. Results of Performance Test (cont'n.)

ITEMS	DATA
9. Average Drying Air Temperature (°C)	97.9
10. Ambient Air Temperature (°C)	
a) Dry Bulb	33.6
b) Wet Bulb	25.5
11. Ambient Air Relative Humidity (%)	52.3
12. Dryer Exhaust Air Temperature (°C)	
a) Dry Bulb	45.6
b) Wet Bulb	42.0
13. Exhaust Air Relative Humidity (%)	80.6
14. Fan Air Velocity (m/s)	11.4
15. Air Flow Rate (m ³ /min)	759.2
16. Plenum Static Pressure (mmH ₂ O)	3.8
17. Power Consumption Rate	
a) Power, kW	11.9
b) Voltage, V	211
c) Current, A	41.5
18. Corn Cob Consumption Rate (kg/h)	125.9
19. Heating System Efficiency (%)	71.4
20. Heat Utilization (kJ/kg moisture removed)	3896
21. Drying Efficiency (%)	60.0
22. Drying System Efficiency (%)	42.9
23. Bulk Density of Dried Granulated Cassava (kg/m ³)	469.4

ANNEX 1. FORMULA USED IN THE COMPUTATION OF DRYING PARAMETERS

$$1. \quad \text{Input Capacity (kg/h)} = \frac{W_1}{T_i}$$

where,

$$\begin{aligned} W_1 &= \text{Initial weight of test material (kg)} \\ T_i &= \text{Input time (h)} \end{aligned}$$

$$2. \quad \text{Output Feed Rate (kg/h)} = \frac{W_o}{T_o}$$

where,

$$\begin{aligned} W_o &= \text{Weight of output (kg)} \\ T_o &= \text{Output time (h)} \end{aligned}$$

$$3. \quad \text{Product Recovery (\%)} =$$

$$P_R = \frac{W_o}{W_i} \times 100$$

$$4. \quad \text{Heating System Efficiency, } E_{HS} (\%) =$$

$$E_{HS} = \frac{Q_s}{Q_f} \times 100$$

Where,

$$\begin{aligned} E_{HS} &= \text{Heating system efficiency (\%)} \\ Q_s &= \text{Heat supplied to the dryer (kJ/h)} \\ Q_f &= \text{Heat available in the fuel (kJ/h)} \end{aligned}$$

To solve for Q_s :

$$Q_s = \frac{\Delta h \times Q}{v} \times 60$$

Where,

$$\begin{aligned} \Delta h &= \text{Change in enthalpy (kJ/kg d.a.)} \\ Q &= \text{Air flow rate (m}^3\text{/min)} \\ v &= \text{Specific volume of fresh air (m}^3\text{/kg d.a.)} \end{aligned}$$

To solve for Q_f :

$$Q_f = F_c \times H_{vf}$$

Where,

F_c = Fuel consumption (kg/h)

H_{vf} = Heating value of the fuel used (kJ/kg)

5. Heat utilization, Q_u (kJ/kg)

$$Q_u = \frac{Q_s \times T_A}{W_1 - W_2}$$

6. Drying Efficiency (%)

$$E_D = \frac{Q_R}{Q_S} \times 100$$

Where,

E_D = Drying efficiency (%)

Q_R = Heat required to vaporize moisture inside the grain (kJ/h)

To solve for Q_R

$$Q_R = Q_v - (W_1 - W_2)/T_A$$

Where,

Q_v = Heat of vaporization of water

7. Drying System Efficiency, E_{DS} (%)

$$E_{DS} = Q_T/Q_f \times 100$$

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N.B.

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